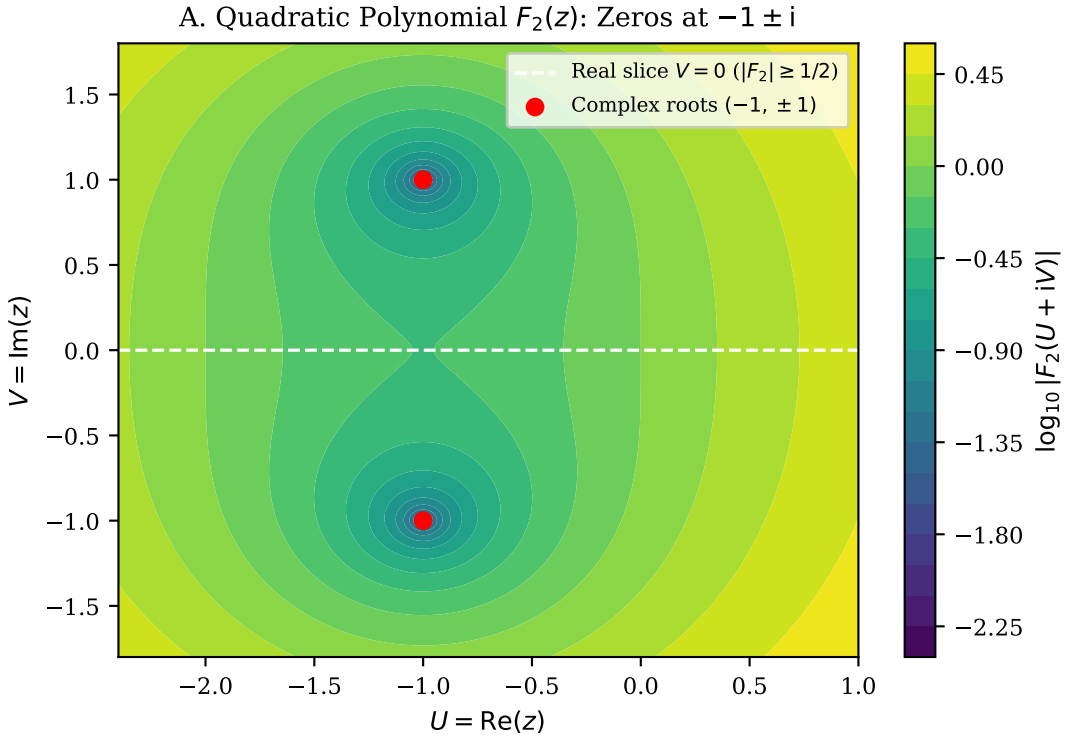
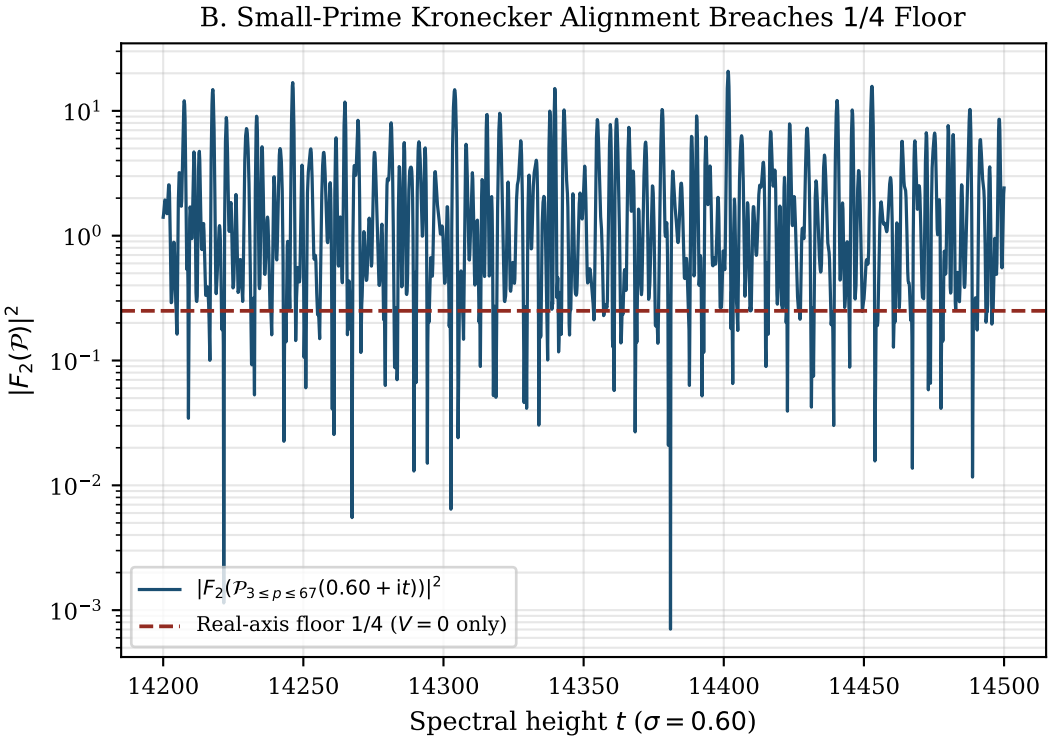
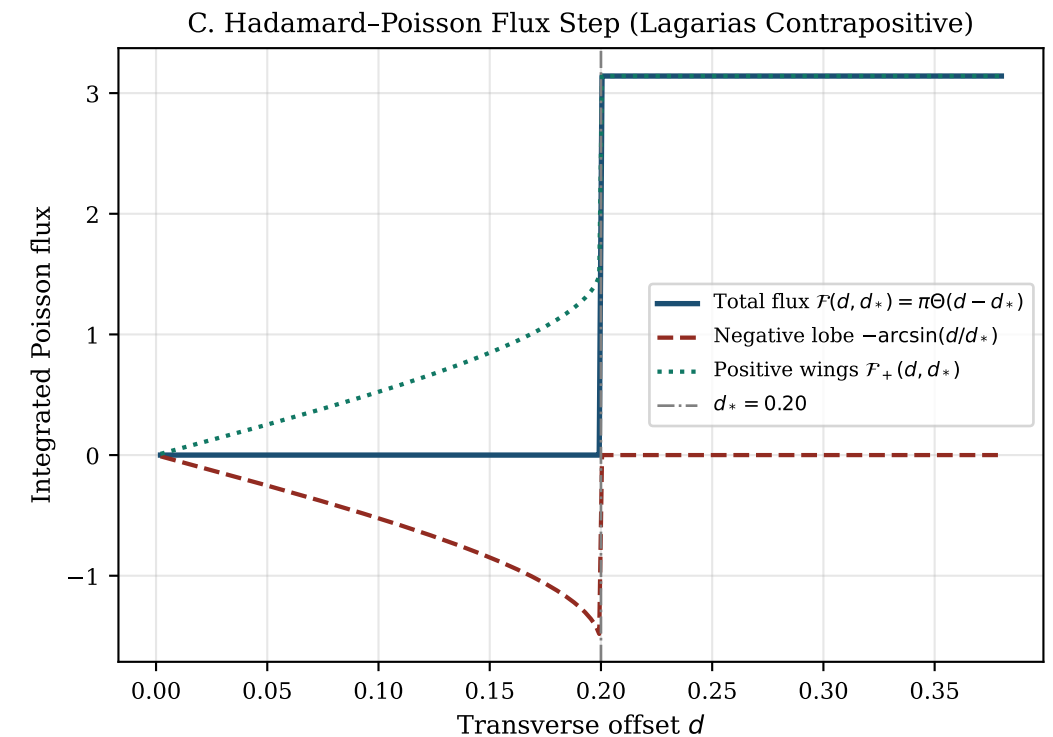
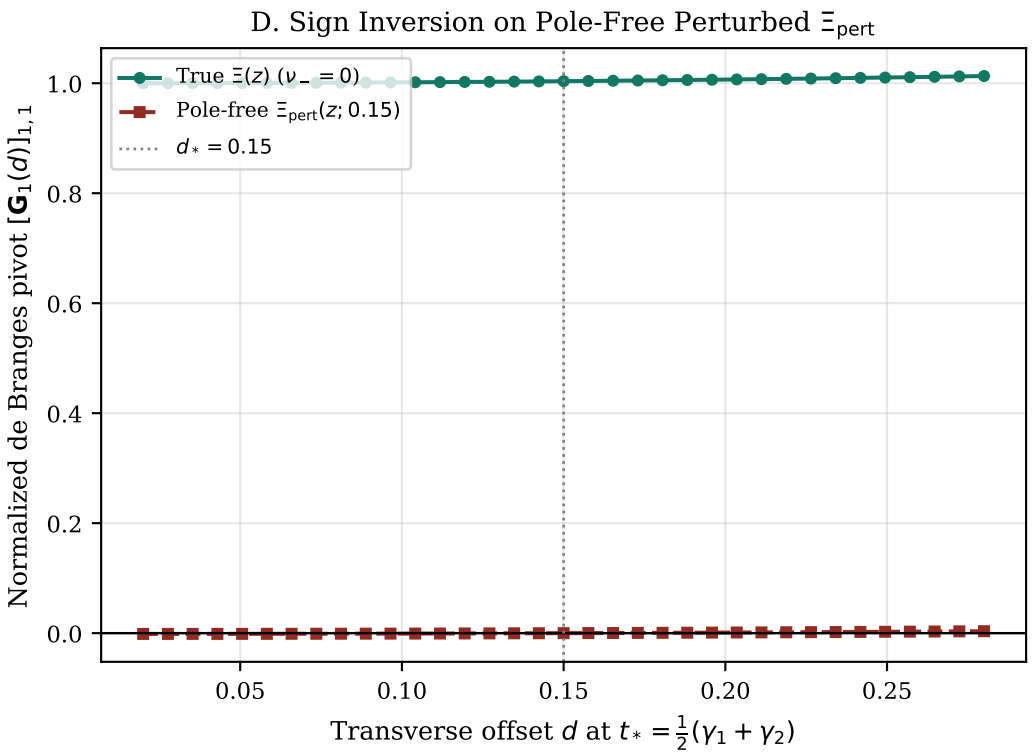

A. Quadratic Polynomial $F_2(z)$: Zeros at $-1 \pm i$
Real slice $V = 0$ ($|F_2| \geq 1/2$)
Complex roots $(-1, \pm 1)$
$V = \mathrm{Im}(z)$
$U = \mathrm{Re}(z)$
$\log_{10} |F_2(U + iV)|$


B. Small-Prime Kronecker Alignment Breaches 1/4 Floor
$|F_2(\mathcal{P})|^2$
$|F_2(\mathcal{P}_{3 \leq p \leq 67}(0.60 + it))|^2$
Real-axis floor 1/4 ($V = 0$ only)
Spectral height t ($\sigma = 0.60$)


C. Hadamard–Poisson Flux Step (Lagarias Contrapositive)
Total flux $\mathcal{F}(d, d_*) = \pi\Theta(d - d_*)$
Negative lobe $-\arcsin(d/d_*)$
Positive wings $\mathcal{F}_+(d, d_*)$
$d_* = 0.20$
Integrated Poisson flux
Transverse offset d


D. Sign Inversion on Pole-Free Perturbed Ξ_{pert}
True $\Xi(z)$ ($\nu_- = 0$)
Pole-free $\Xi_{\mathrm{pert}}(z; 0.15)$
$d_* = 0.15$
Normalized de Branges pivot $[\mathbf{G}_1(d)]_{1,1}$
Transverse offset d at $t_* = \frac{1}{2}(\gamma_1 + \gamma_2)$